

The Future of Quantum Artificial Intelligence

Abstract: This paper explores the intersection of quantum computing and neural networks. Quantum AI promises exponential speedups for complex optimization problems and machine learning training phases. By leveraging quantum superposition and entanglement, deep learning models can potentially evaluate multiple states simultaneously, drastically reducing the time required to process large datasets.

1. **Introduction:** Traditional computing architectures are reaching their physical limits. Quantum algorithms, such as Grover's and Shor's, have already demonstrated theoretical superiority. When applied to AI, parameterized quantum circuits (PQCs) can serve as trainable models akin to classical neural networks.